Procedural Shapes and a Moving Policy Rollout

Technical Report

Submission ID: Anonymous submission

Abstract: Episode 38 is admitted. Our Track 3 entry uses its learned moving rollout and keeps v5's static poses on nineteen episodes. The final pack follows this report. One combined file supplies meshes and poses to all six boards. Pretrained shape gains remain development findings; scan reuse and amended moving gates limit the evidence. All counted-file scores remain pending.

1. Introduction

Track 3 scores video-derived meshes and simulated policy rollouts [2, 3]. We combine MoGe-2 [5], procedural shapes and FlashCHORD [1]; Hunyuan3D-2mini [4] supplied development shapes. AI coding agents led and implemented the work under a human owner's direction. We report a traced combined entry, scan-informed shape gains that did not reach the final file, and object-path tracking without recovered human hands. The evidence does not establish transfer or dexterity.

2. Approach

Counted shapes and poses. We pooled above-table MoGe extents and manually specified parts, excluding merged/clipped observations. Public scans informed design and selection, but no scan geometry, dimensions or templates enter meshes or simulated assets. Static RL policies use boxes, inactive hands and 512-step budgets, with learned checkpoints exported through the unedited evaluator: this does not demonstrate learned manipulation. Stereo mask centroids place box 2 with 1 cm clearance; merged masks retain earlier shrinkage. Calibration transfer is assumed; unknown heading rotates the aligned second-object direction.

File and container. The kit change forced repacking after the planned noon cut-off. The entry keeps nineteen v5 exports within six-decimal CSV rounding (maximum 4.98×10^{-7}). Meshes use rollout body frames, with 31 shared coordinate maps, two recorded rotations on 19, and pot 5/pitcher 20's procedural v4 shapes from the same constructor; their pretrained shapes have no recorded map. Shared coordinates do not imply equal surfaces. Pooled D1 dimensions are retained, not resized to simulation bounds. The same single file goes to all six boards, so pose and CD-O shapes are identical. 38 uses its simulated rollout mesh instead of the static pan's procedural shape, changing CD-O by an unmeasured amount. The container runs nineteen static policies and 38's learned moving policy through the unedited evaluator with these meshes.

Moving experiment. Fixed SAM2.1/Grounding DINO prompts select the smallest box above a fixed threshold; highest score selected 38's table. Background KLT bundle adjustment estimates camera motion; stereo and silhouettes track our video-derived pan. Per-frame tracking uses no scan or mocap; one public-scan constant sets bounding axes, origin and handle convention on both clips, creates no motion and enters no mesh/asset. A 2 cm overlay ring offset is unresolved. FoundationPose is not in the entry. FlashSAC trains from scratch with 1,024 worlds, seed 0, update ratio 2, a compressed curriculum and 66,763,776 transitions; final assistance is off. A fixed rigid grasp replaces recovered human hands. Fingers stay fixed, and video hands start in place without prepended rest. Contact alone moves the object. Hand–support collisions are off; mass 0.3 kg and diagonal inertia 0.001 kg m^2 are example values, not measured. A thin support lies below the hulls. The shared simulated profile is oversized (25.6 by 21.3 versus 18.2 by 16.6 cm), altering grasp/contact.

3. Experiments and Results

Shape findings. G2's pretrained pot 5/pitcher 20 shapes were measured but did not reach the final file. Packed equal-object/pair CD-O fell 0.6534/0.6629 to 0.5644/0.6208 cm on ten reused public scans, 26/36 entries; ten spoon entries remain unmeasured. Adaptive scan selection prevents new-object validation. Admission is unvalidated; the packed veto supported the gain, mostly pitcher. Generation varies by seed; pot 5 was a favourable draw. G3 and silhouette vetoes failed; fusion stopped before scoring.

Static findings, old scorer. Eight equal-weight train episodes, one seed: AUC 0.2690 to 0.2767, success rates zero; one episode gives the gain, four changed episodes retain 51–130° direction errors. The shrinkage control does not isolate calibration from shorter offsets. Selection used train initial-frame errors despite incomplete fresh confirmation. Simulation consistency does not measure mocap accuracy. V5 changes heading only on 19 via scan-

to-stereo registration, supported by train 18 alone (old AUC 0.30 to 0.47, one seed). Its pre-written selected/flipped stress rule passed, conditional on transfer, not calibrated heading probability; a wrong basin can hurt 19. Even true initial poses held static reach only old-scorer train AUC 0.532.

Both scorers. The organisers replaced the kit on 2 November; every earlier local pose number used the old scorer. New-scorer *public-geometry diagnostics*, not host predictions: static train AUC 0.302 (0.292/0.293 at other registration seeds) versus old 0.2767, on frozen kinematic projections, not eight train RL exports. On 39's real video-policy export: 0.669 with the simulated mesh, 0.591 with the tracking mesh, versus old 0.775; MP-SR falls 1 to 0. Public visual scans, centroid anchors, 500 sampled vertices and assumed PCA axes substitute for unpublished host geometry/anchors/axes/points. Primitive registration flips up to 179.93°, making body rotation arbitrary. These diagnostics changed no method, gate or threshold.

39 path (old scorer)	AUC	E cm	angle deg
Held still	0.18	16.43	51.08
Video path alone	0.833	2.16	8.57
Policy on mocap	0.982	0.62	0.92
Policy on video	0.775	2.67	8.31

Table 1: Tuning 39, single first-pose alignment to mocap, old scorer. E/angle: mean translation/rotation error. One sealed tracker label pass; the path is not a rollout, mocap training is an upper bound. 38's ground-truth accuracy is **unknown**.

38's self-reference $(T, Q, E, \Theta) = (1.026, 0.973, 1.42\,\text{cm}, 1.59°)$ passes $(\geq 0.50, \geq 0.50, \leq 3\,\text{cm}, \leq 15°)$; T is displacement gain; Q is one minus normalised squared error of 15-frame position increments (zero held still). The policy completed all 263 frames without a reset. The first pose 0.30 cm/0.27° passes 1 cm/2°; mask IoU median/p10/coverage 0.778/0.707/100% passes 0.60/0.30/90%. These old-protocol gates show goal following, not truth or superiority to static v5. 38 passed at 12:58 UTC, after our planned noon cut-off, which the owner then moved (about 13:25 UTC); admission passes and packing follows this report.

Across three attempts, 38 took at least 63 minutes. The second record is stale; the bound uses its last log write. The 34.3–40.4M stretch was kept. The same method on spoon clips 35, 36 and 37 stopped at the mask-coverage check; all three keep their static exports. Usable-mask fractions were 10.9, 8.8 and 0.7% against the 90% rule.

Integrity. The tracker failed its frozen rule and amendment. A conditional third rule moved pixel limits/instruments after seeing both clips' video gates, before one sealed same-object mocap pass on 39, breaking "only one amendment". Perpendicular residual required passing labels; 38 required defective table checks fixed at unchanged limits, then fresh withheld tracks passed. This is weaker than prospective validation. A constant-frame bug was fixed before labels. Camera error cancels in reprojection; silhouette fit partly measures its own objective. Withheld table/long-baseline landmarks address some drift, not object truth. M2v's NULL became GO after retrospective first-pose alignment, unspecified in its protocol; the NULL remains, labels were read twice for arithmetic, though the policy was scored once. Alignment hides initial placement/heading, not drift, scale or timing. 38 failed midpoint $T = 0.25$, cause undiagnosed; two stops (kill rule, owner) lost replay twice. A resume disallowed by the kill rule was necessary; this differs from 39's fresh run. One object/seed, a non-human hold and unscored contact establish neither transfer nor dexterity.

Leaderboard. D1 v5 scored 0.53562 cm but is not the counted CD-O entry. Counted-file CD-O: [pending]; AUC: [pending]; SP-SR: [pending]; MP-SR: [pending]; RPE: [pending]; MPPE: [pending]. Scores came after every method choice and steered nothing.

4. Conclusion

38 has no labels; its accuracy is unknown and nothing shows it beats its static export. On 39 (one clip/seed), new-scorer public-geometry AUC is 0.175 held still and 0.669 moving. **Industry Innovation (one moving episode; nineteen static). Economic Viability:** On 39, 66.76M transitions took 58–62 minutes per policy on one 16 GB GPU (sampled peak 11.4 GiB); cost at scale is unshown (one object/seed, 38 restarted twice). **Data Flywheel:** No flywheel is shown: two clips reuse one hashed asset and video-only gates frozen after revisions, but nothing improves itself over time. **Complexity Reduction:** Stereo video, fixed prompts and our shape with rigid-grasp hands simplify inputs; a scan constant, two failed rules and one sealed label pass leave deployment savings unshown. **Innovation:** Testing whether recovered human hands are necessary, a video-path policy reached old AUC 0.775 versus held-still 0.18 (new public-geometry 0.669/0.591), with a non-human hold, oversized stand-in and one tuning clip; this method is in the counted entry on 38, while nineteen episodes stay static.

References

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[3] NVIDIA. Video to Data Challenge 2026: Frequently asked questions, 2026. URL https://nvidia-isaac.github.io/video_to_data/v2d_challenge/assets/v2d_challenge_2026_faqs.pdf.

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